

ChangeLoop

Salt sets the coal bill, not water. A decision layer for zero-discharge dyehouses, and what it finds when four machines share one evaporator.

In one paragraph

In a dyehouse running zero liquid discharge, the coal bill is set by how much salt goes into the dye bath, not by how much water comes out. That single fact is counter-intuitive, it is provable from mass conservation, and almost nobody acts on it. It means most water-saving work in India's largest textile cluster is aimed at a number that does not drive the emissions. ChangeLoop puts the downstream thermal consequence in front of the person making the upstream scheduling decision, at the moment they make it. It is live, it is open to inspection, and every figure in it carries a label saying how much it should be trusted.

Live prototype: <https://changeloop-water-intelligence.onrender.com>

1. Problem Statement

The place

Tirupur, in the Noyyal sub-basin of the Cauvery system, is India's knitwear capital. In 2011 the Madras High Court ordered roughly 700 dyeing and bleaching units shut for discharging saline effluent into the Noyyal. They were allowed to reopen only under zero liquid discharge: nothing leaves the site as liquid.

ZLD worked. The river stopped receiving effluent. But it moved the problem rather than removing it, and the place it moved to is the boiler house.

What ZLD actually does

Under ZLD, effluent goes through membranes. The clean permeate returns to the process. The concentrated reject goes to a multiple-effect evaporator, which boils it until only solid salt remains. Boiling water takes heat, the heat comes from steam, and in Tirupur the steam comes overwhelmingly from coal.

So a regulation written to protect a river created a large, permanent thermal load. Evaporating one cubic metre of reject takes about 188 kWh of thermal energy. Latent heat of 0.627 kWh/kg from steam tables, divided by a steam economy of 3.33 kg of water per kg of live steam, which is the reciprocal of the 0.25-0.35 kg steam per kg evaporated that textile MEE trains are reported to achieve. The result lands inside the 150-250 kWh/m³ band independently published for the evaporator section of textile ZLD plants, which is a useful check: two unrelated published numbers, multiplied, fall inside a third that was never used in the calculation.

The mechanism nobody is looking at

Here is the part that matters, and it is the reason this project exists.

In a closed loop, the volume the evaporator has to boil is not set by how much water you used. It is set by how much salt you put in:

$$V_{\text{reject}} = M_{\text{salt}} / C_{\text{reject_max}}$$

Salt mass divided by the maximum concentration the membranes can push the reject to, which for a textile RO train is around 60,000 mg/L. Water entering the loop comes back round. Salt does not. Salt has to

leave as a solid, and the only exit is through the evaporator, which means the evaporator must boil whatever volume of water that salt is dissolved in at the concentration ceiling.

The consequence is sharp enough to be tested, and the running system tests it on every request:

Intervention	Effect on evaporator energy
Cut effluent water volume by 20%	+0.0%
Cut salt load by 20%	-20.0%

Not "a little". Zero, to the precision of the arithmetic. It falls directly out of salt mass conservation.

This is why a water dashboard cannot find this saving. Two interventions that both read as "saving water" on any volume meter do completely different things to the coal bill, and nothing on the factory floor distinguishes them.

Why it has stayed unsolved

The decision that sets salt load is made by a production planner, choosing what order to run today's dye lots in and which rinse strategy to use. They make it hours before anything is visible. The evaporator's fuel bill arrives weeks later, on a different cost centre, with no way to trace it back to the decision that caused it.

That is a broken feedback loop, not a technology gap. The planner is not careless. They are working without the one number that would change their mind, and no system currently puts it in front of them.

And one layer above that

There is a second failure, and it only appears when you stop looking at one machine.

A real mill runs several machines into one shared evaporator. Each planner picks the plan that is best for their own machine. Every one of those choices is locally correct. Their sum can still exceed what the shared plant is able to boil.

In the modelled site, four machines each choosing their own optimum put the shared evaporator at **121.9% of capacity**. Nobody chose badly. The sum does not fit. No tool that looks at one machine at a time can see this, and no planner can be blamed for it.

2. Proposed Solution

ChangeLoop prices the downstream consequence into the upstream decision.

It is a decision-support layer that sits where the schedule is made. Before a shift starts, it searches the combinations of lot order and process strategy available to the planner, and for each one computes the full consequence chain: water demand, salt load, reject volume, evaporator steam, CO₂e, rupees, and whether any ship date is missed.

Four design choices do the actual work.

Hard constraints are feasibility, not penalty. A plan that misses a ship date is not a worse plan. It is not a plan. It is removed from the search rather than given a bad score, because a schedule that saves water by shipping late is not a trade-off a factory can accept and should never be offered as one.

Every coefficient is classified and visible. Each of the 25 numbers that drive the model is labelled PUBLISHED, DERIVED, ASSUMED or MEASURED, with its source and its arithmetic. Anyone can open the registry in the running system and check. Currently: 17 published, 4 derived, 4 assumed, and **zero measured** — a test enforces that last count so it cannot quietly become a marketing claim.

The system fails closed. It issues no setpoints. It cannot release a dye bath. Early wash-off release requires a named human to approve it, and `automatic_release` is hardcoded to `False`. If a sensor reading is missing, frozen or implausible, the system holds and credits zero saving rather than estimating. An advisory tool that quietly guesses is worse than no tool.

Savings are credited only when a human approved the action. If a recommendation is not approved, it contributes zero to the ledger, even though the engine computed it. The ledger records the decision actually taken, not the decision that would have looked best.

The plant layer

The newest component answers the coordination failure described above. It compares two plans over the same day: what each machine would choose alone, and the cheapest combination that actually fits the shared evaporator.

The capacity figure is **derived, not invented**. In a closed loop the freshwater makeup is the evaporative loss is the reject volume, so the site's daily abstraction allowance and the volume the evaporator must boil are the same number read two ways. This matters, because the easiest way to make a coordination demo impressive is to pick a capacity that guarantees a breach. That number is the basin allowance the rest of the system already runs on.

And the result is the most interesting finding in the project:

Binding constraint	Plant load	Coordination needed	Premium
Normal operation	121.9%	Yes — 2 of 4 machines forced off their own optimum	₹4,123/day
Scarcity priced (drought)	68.8%	None	₹0

Under today's tariffs, freshwater is cheap enough that every machine privately prefers the water-saving plan, and their sum breaks the evaporator. Price scarcity properly, and all four machines choose the low-salt plan on their own account. The plant fits, nobody is made worse off, and the premium disappears.

Nothing was rescheduled to get there. **The breach is a pricing failure, not a scheduling failure.** The plan the site has to force onto two machines today is the plan every machine would pick for itself if water cost what it is worth. That is a policy argument that comes out of the engine rather than out of an opinion, and a test pins it so it cannot drift.

3. Prototype Details

A working system, publicly reachable, not a mockup.

<https://changeloop-water-intelligence.onrender.com>

Backend	Python standard library only — no framework, no install step
API	34 REST endpoints, every impact figure computed server side
Tests	178, covering physics, safety interlocks, concurrency, switching-price search and the UI
Front end	Dependency-free; ten operating views

Backend	Python standard library only — no framework, no install step
Ledger	HMAC-SHA256 append-only hash chain, verified on every health check

What you can do in it

Choose a site and a binding constraint. Watch the optimiser change its recommendation as the constraint changes — it returns two genuinely different plans across four modes, which is the test of whether it is optimising or just replaying an answer. Open any number and trace it to its coefficient, its source classification and its arithmetic. Approve a sequence, or decline it, and watch the ledger credit accordingly. Trigger a sensor fault and watch the system hold instead of guessing. Open the Plant view and see four correct decisions produce one breach.

How it is validated

Nothing here has been measured on a real asset, and the system says so in every view. What it has instead is external cross-validation.

Fed the inlet TDS that CPCB measured at an assessed Tirupur dyeing unit — 18,340 mg/L — the model predicts a reject fraction of **30.6%**. Indian textile ZLD operators independently report RO reject at 20-30% of inlet volume. The model lands inside that band using an input it never reads back, and the check runs as a test rather than sitting in a slide.

The honest reading of that: the physics is behaving like real plants. It is not evidence that any specific figure describes any specific site. Those are different claims and the system keeps them apart.

Deliberate scope limits, stated in the product

The interface carries its own limitations rather than hiding them: no scheduling of machines against each other in time, no surge storage modelling, machines within one unit rather than a CETP's hundreds of members, a simulated order book, and nothing measured anywhere.

4. Target Audience

Who uses it

The production planner is the primary user. They already make this decision every shift; ChangeLoop changes what they can see when they make it. This matters for adoption — the tool does not ask anyone to do a new job.

The quality supervisor holds authority over wash-off release and keeps it. The system can recommend early release; only they can grant it.

The unit owner takes the financial benefit and signs off the chemistry decision, which runs on a procurement cycle rather than a daily one.

Who buys it

The common effluent treatment plant is the efficient channel. One connection reaches many member units, and the incentive already points the right way: less salt arriving means less steam bought and less solid salt to store. India's textile ZLD plants are holding over one lakh tonnes of recovered salt with no viable disposal route, so reducing salt at source is something a CETP operator wants for reasons that have nothing to do with us.

Who benefits without using it

Households and farmers in the Noyyal sub-basin, whose groundwater carries a basin stress weight of 11.08 in this model — every litre not abstracted here counts for far more than a litre elsewhere. And the cluster's workforce, for whom the 2011 shutdown is living memory rather than history.

5. Impact and Feasibility

What one shift looks like

Baseline is a conventional single-stage rinse. Everything below is MODELLED on a simulated order book of five lots.

	Freshwater	Evaporator steam	Cost
Baseline	11,907 L	2,239 kWh	₹33,897
Recommended under normal economics	11,187 L (-6%)	2,104 kWh (-6%)	₹26,038
Available under priced scarcity	6,369 L (-47%)	1,198 kWh (-47%)	₹29,984

That 6% against 47% gap is the most important number in this document, and it is not a weakness in the optimiser. The 47% plan is fully feasible — it misses no ship date. It is not selected under normal operation because the low-salt chemistry premium costs more than the steam it saves at today's tariffs. The engine is telling the truth about the economics it was given.

Which is precisely the finding. **The large saving is available and nobody is buying it, because the price signal does not reach the person deciding.** Price water and carbon properly and the same engine, unchanged, recommends the 47% plan. We did not hide this behind the better number.

What price would change the answer

Saying "price water properly" is easy. The engine computes what that would actually take. It walks each price across a range, re-running the full optimiser at every step, and finds the first value at which a profit-seeking mill picks the 47% plan on its own account.

Price	Today	Switches at	Change needed
Low-salt dye premium	Rs 6.50/kg fabric	Rs 5.74/kg fabric	-11.6%
Boiler steam cost	Rs 2.03/kWh thermal	Rs 3.13/kWh thermal	+53.9%
Freshwater tariff	Rs 45/m3	Rs 251/m3	+457%

The dye premium needs to fall by under 12%. That is the distance between the plan the cluster runs today and a plan that cuts freshwater and evaporator steam nearly in half. It is a procurement question, not a research question.

The steam lever converts straight into a carbon price. Steam would have to rise by Rs 1.09 per kWh of heat; delivering that heat emits 0.452 kg of CO₂e; so the carbon price that produces the rise is **Rs 2,423 per tonne, about EUR 21.**

The EU Emissions Trading System was charging EUR 82.40 a tonne on 5 October 2026, roughly four times more. This abatement is not waiting on an unreachable price. It is waiting on any price at all.

Two caveats travel with that figure and the system states both. It assumes the whole carbon cost reaches the mill as a higher steam price, which is the optimistic case; where it does not, the price needed is higher. And it is the level at which this particular decision flips in this modelled mill, not a proposal for what a carbon price should be.

At cluster scale

A linear projection across 400 units, 900 lots each per year. This is a PROJECTION and the system labels it as one — it assumes every unit resembles the modelled one and that every recommendation is approved, neither of which has been tested.

Freshwater avoided	75 million litres/year
Salt avoided	4,500 tonnes/year
Evaporator steam avoided	14,106 MWh/year
CO2e avoided	6,376 tonnes/year

Freshwater avoided equals reject avoided exactly, because in a closed loop the makeup water is the water that was evaporated. That identity is enforced by a test, after an earlier version of this projection violated it and overstated water by a factor of five. Finding and fixing that is part of why the identity is now a test rather than a convention.

Why it is feasible

No capital equipment. The intervention is a decision, not a retrofit. The first two phases change nothing on the floor at all.

It rides an existing mandate. ZLD is already compulsory and already paid for. ChangeLoop makes existing assets cheaper to run; it does not ask anyone to install one.

Advisory from the start. The existing planning procedure stays authoritative throughout. That is what makes a pilot approvable.

A 32-week staged pilot, defined in the engine rather than written for this document: baseline metering, then shadow mode where recommendations are produced and nobody acts, then advisory with approval, then a controlled wash-off release trial, then the chemistry decision, then expansion. Alternate weeks on and off, so the comparison is against the same season, order mix and crew.

The pilot is designed to answer one question: *did ChangeLoop cause the reduction, or would it have happened anyway?* A single wash-fastness failure stops the release trial, because a re-processed lot costs more water than was saved.

What would falsify this

Worth stating plainly. If a pilot measures evaporator steam per cubic metre far from 188 kWh, or finds the 60,000 mg/L concentration ceiling does not hold across a real operating range, the magnitude of everything here changes. The direction would survive — it follows from mass conservation — but the numbers would not. That is exactly what phase 1 exists to find out, and better to know in week six than after a cluster has paid for it.

6. Pitch and Idea Diffusion

The pitch, in three sentences

Zero liquid discharge saved the Noyyal and handed the cluster a coal bill. That bill is set by salt, not water, so most water-saving work in Tirupur is aimed at the wrong number. ChangeLoop shows the planner the real consequence at the moment they choose, and the engine proves the claim rather than asserting it.

How the idea spreads

Through the CETP, not unit by unit. One commercial relationship reaches many members, and the CETP's own economics already favour less salt arriving.

Through shadow mode. Four weeks of being correct while changing nothing is how a tool earns the right to influence a factory's decisions. It is the cheapest trust-building mechanism available and it costs the site nothing.

Through the open coefficient registry. Every number, source and derivation is inspectable in the running system. A process engineer who disagrees with a coefficient can see exactly what it is and what it changes. That is a faster route to credibility in an engineering community than any amount of marketing, and it invites the correction rather than defending against it.

Through the pricing argument, with a number attached. Where a shared environmental resource is under-priced, individually rational decisions overload it, and the fix is a price rather than a schedule. Most submissions stop at that sentence. This one names the price: about EUR 21 a tonne of CO₂e, or a 12% fall in the dye premium. A tariff-setter can act on a number and cannot act on a principle, and the engine behind it is open for anyone to re-run.

Through being wrong in public. This submission names its own largest gaps. That is a diffusion strategy as much as an honesty one — in a field full of unverifiable impact claims, the system that publishes its own weak points is the one an engineer will actually try.

7. Future Scope

Near term — make it measured

The single biggest upgrade is two instruments: evaporator steam flow, and reject conductivity. Those two readings promote the two coefficients carrying the most weight in the result from PUBLISHED to MEASURED. Nothing else moves the credibility of this system as far for as little. They also need calibration and health monitoring, because a drifting sensor is worse than no sensor, and the existing rule — missing or implausible data means hold and credit zero — has to survive scale rather than being relaxed for throughput.

Medium term — the cluster

The current plant solver is exhaustive. It optimises each machine's queue within every strategy, then checks all 256 combinations against capacity and returns the cheapest that fits, so the answer is proven rather than approximated. That stops being possible at a CETP with hundreds of member units, and the method has to change. This is a real limit and not a detail: cluster-scale coordination is a different optimisation problem, not a bigger version of the same one.

The commercial question underneath it is harder than the mathematics. If the cheapest plant-wide plan requires one unit to accept a worse outcome, somebody has to pay them. The engine can already compute what that transfer should be — ₹4,123 a day in the modelled site, attributable per machine. Whether a CETP can actually administer it is an institutional question, and it is open.

Longer term — the same identity elsewhere

The physics ChangeLoop runs on is not specific to textiles. Any closed-loop water system that concentrates dissolved solids and removes them thermally obeys the same relation: the volume you must evaporate is set by the dissolved mass divided by the concentration ceiling, not by the water you circulated.

Evaporative data centre cooling has exactly this shape. Makeup water equals evaporation plus blowdown; blowdown volume is set by the cycles of concentration the water chemistry permits; and cycles of concentration are limited by dissolved solids. The same structural blind spot should therefore exist — operators optimising water withdrawal while the dissolved-solids ceiling quietly sets the thermal and chemical load.

Stated carefully: that is a structural argument, not a validated finding. No data centre has run this and the coefficients would all need re-sourcing. Cooling towers, boiler blowdown and mine water management share the same structure. It would be dishonest to present any of them as solved, and the textile case has to be proven on a real site first.

Sources

The authoritative, per-coefficient sourcing is not in this document. It is in the running system, where every one of the 25 coefficients carries its own source text and arithmetic and can be read without asking us: <https://changeloop-water-intelligence.onrender.com/api/evidence>

That is deliberate. A bibliography in a concept note can drift from the model; a registry the model reads at runtime cannot.

The external reading behind the figures quoted above:

Figure	Source
Grid emission factor, 0.705 kg CO ₂ e/kWh	CO ₂ Baseline Database for the Indian Power Sector, Central Electricity Authority (cea.nic.in). The registry uses the version current at the time of writing.
Carbon price benchmark, EUR 82.40/tonne, 5 October 2026	European carbon prices, S&P Global Commodity Insights. A market price, so it goes stale; the system stores it with its date.
Tirupur ZLD mandate, CETP capacity and water recovery	"Towards zero discharge", Down To Earth.
Tirupur cluster effluent load and disposal practice	"Study of Tirupur textile industry cluster", India Water Portal.
Inlet TDS of 18,340 mg/L; RO reject at 20-30% of inlet; MEE steam use of 0.25-0.35 kg steam per kg evaporated	Central Pollution Control Board assessment of textile dyeing units and ZLD at Tirupur. Quoted in full in the registry entry for each coefficient.
Water and steam tariffs	Tamil Nadu Electricity Regulatory Commission tariff orders and Tamil Nadu pollution board plant data, as cited per coefficient in the registry.
Latent heat of vaporisation, 0.62694 kWh/kg	Standard steam tables: 2,257 kJ/kg at 100 C and 1 atm, divided by 3,600.
Evaporator section energy band, 150-250 kWh/m ³	EU Best Available Techniques reference document for the textiles industry, used only as an independent check and never as an input.

Where a source gives a range, the registry stores the range and the model takes a stated point inside it. Where public literature could not settle a figure, it is labelled ASSUMED rather than attributed to a source that does not say it. Four coefficients are in that state and all four are commercial prices.

Closing

ChangeLoop is a working system, publicly reachable, with 178 tests, 25 classified coefficients, external validation against CPCB measurements, and zero measured data — and it tells you that last part itself, in every view.

The core claim is small enough to check and large enough to matter: in a closed loop, salt sets the coal bill. The person who controls salt is a planner who cannot see the consequence. We built the thing that shows them, proved the physics, priced the gap, and found that at plant scale the problem is not that anyone is deciding badly — it is that water is too cheap for good decisions to add up.

<https://changeloop-water-intelligence.onrender.com>